

RAW-2-COMPASS

App download link: <https://compassprep.precsn.com/>

About this guide

This guide walks through downloading and opening the prebuilt macOS app, choosing FASTQ files, running the analysis, and finding the output files. Screenshots from the app are included where available.

Important: app download and source code are different

The GitHub project page is: <https://github.com/poojabio/RAW-2-COMPASS>

The GitHub Releases page is: <https://github.com/poojabio/RAW-2-COMPASS/releases>

The repository provides downloadable source code, but the source-code ZIP is not the ready-to-open app.

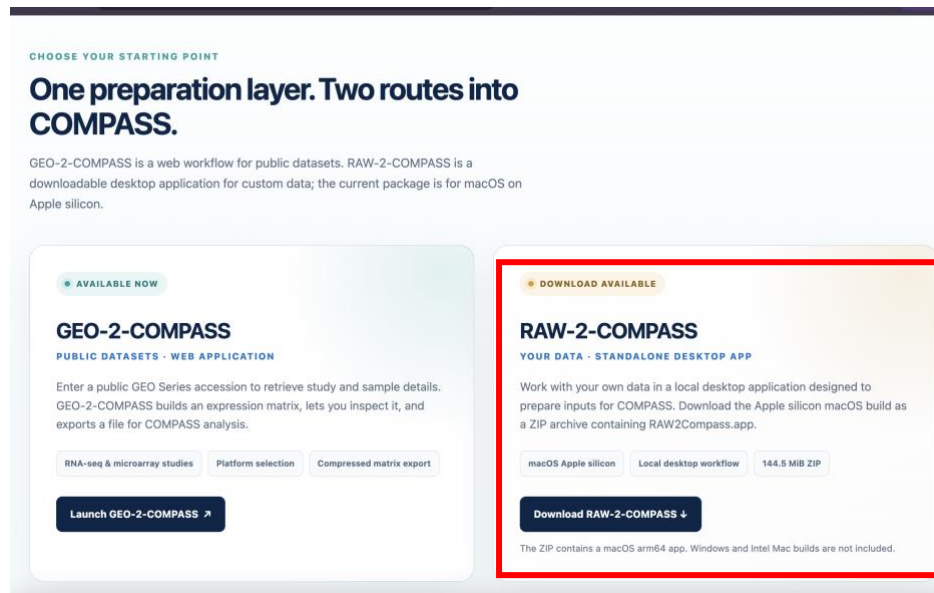
To use the app without setting up Python, obtain the prebuilt macOS app ZIP from the distribution link supplied by the project maintainer:

<https://compassprep.precsn.com/>

The current bundled app is intended for Apple silicon Macs (arm64). Confirm that the download you received is compatible with your Mac.

1. Download the app

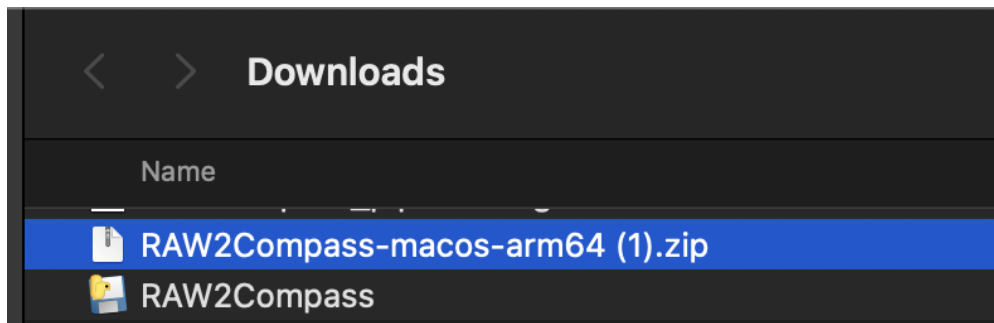
1. Open the app download link supplied by the project maintainer.
2. Download the macOS app ZIP file to your Mac, such as the Downloads folder.
3. Wait for the download to finish. The downloaded ZIP contains the app bundle.



Screenshot 1: Download page showing the correct macOS app ZIP (RAW2Compass-macos-arm64.zip).

Move to Trash).

4. Extract the ZIP first. In Finder, open Downloads and double-click **RAW-2-COMPASS-macos-arm64.zip** so the RAW2Compass app appears next to it (details in Step 2).
5. In Finder, open Downloads (or the folder where the ZIP was saved).
6. Double-click the ZIP file. macOS will extract an app bundle ending in ".app".
7. Double-click the app to open it.
8. You may rename the app bundle in Finder if desired. For example, select it, press Return, type "RAW-2-COMPASS", and press Return. Keep the ".app" extension if Finder displays it.



Screenshot 1b: First launch of the app.

2. Allow macOS to open the app (required for now)

Why this step is needed. RAW-2-COMPASS is unlicensed software right now: it has not been signed and verified by Apple. Because its downloaded from the internet, macOS marks

it as quarantined, showing the message below. One Terminal command removes that quarantine flag from this one app so it can open. Do this once, right after downloading.



Screenshot 2a: The macOS message shown when the app is opened before this step. Click Done (do not click

9. Open Terminal: press **Command + Space**, type **terminal**, and press Return.



Screenshot 2b: Opening Terminal with Spotlight (Command + Space).

10. Type the command below, replacing **YOUR-USERNAME** with your Mac username, then press Return:

```
xattr -rd com.apple.quarantine /Users/YOUR-USERNAME/Downloads/RAW2Compass.app
```

```
xattr -rd com.apple.quarantine /Users/poojaparthasarathy/Downloads/RAW2Compass.app
```

Screenshot 2c: Terminal command demonstration

Make sure the path matches where your app is

Easiest way: type `xattr -rd com.apple.quarantine` followed by one space, then drag the app icon from Finder into the Terminal window. Terminal fills in the full path for you. Press Return.

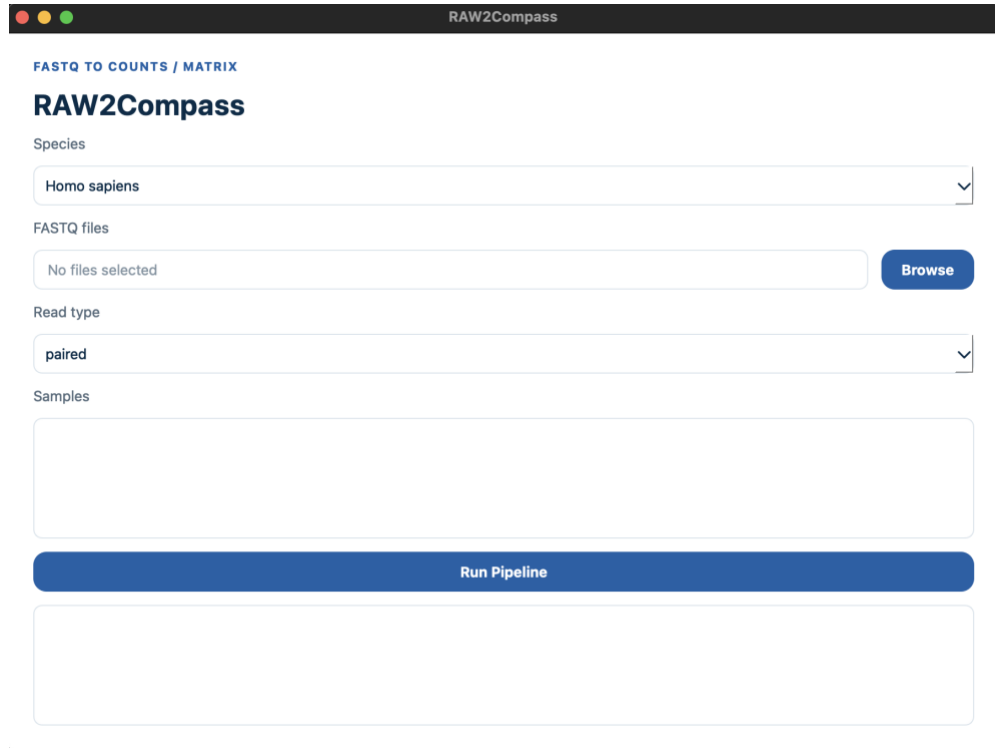
`/Users/` is the start of your home folder, followed by your Mac username. Your username is the name before the `@` in the Terminal prompt.

`Downloads/` is the folder where the app was downloaded and extracted. If you moved the app, for example to Applications, use that location instead, such as `/Applications/RAW2Compass.app`.

`RAW2Compass.app` is the app name. If you renamed it, use the new name.

Shortcut: ~/Downloads/RAW2Compass.app means the same as /Users/YOUR-USERNAME/Downloads/RAW2Compass.app.

1. If nothing prints and a new prompt appears, it worked. Now open the app by double-clicking it. (Example of a file path from a user computer)

The screenshot shows the RAW2Compass application window. At the top, the title bar says "RAW2Compass". Below it, the text "FASTQ TO COUNTS / MATRIX" is displayed. The main heading is "RAW2Compass". There are four input sections: "Species" with a dropdown menu showing "Homo sapiens"; "FASTQ files" with a text box saying "No files selected" and a blue "Browse" button; "Read type" with a dropdown menu showing "paired"; and "Samples" with a large empty text area. At the bottom, there is a blue "Run Pipeline" button and another empty text area below it.

Screenshot 2d: First launch of the app.

3. Choose species, read type, and files

1. In the app, choose the reference species:
 - Homo sapiens (human)
 - Mus musculus (mouse)
 - Canis lupus familiaris (dog)
 - Macaca mulatta (rhesus macaque)
 - Danio rerio (zebrafish)
2. Choose the read type: "paired" or "single".
3. Choose the read type **before** clicking Browse. The app determines the sample list when files are selected.
4. Click Browse and select the FASTQ file(s).
 - For paired-end data, select both reads for every sample in the same selection.
 - For single-end data, select each file to process as a sample.
5. Review the Samples list. Detected samples are checked by default. Uncheck any samples you do not want to process.

6. Check the log area. In paired mode, files that could not be matched are skipped and the app reports how many were skipped. Make sure every intended sample appears before continuing.

FASTQ TO COUNTS / MATRIX

RAW2Compass

Species

Homo sapiens

FASTQ files

No files selected **Browse**

Read type

paired

Samples

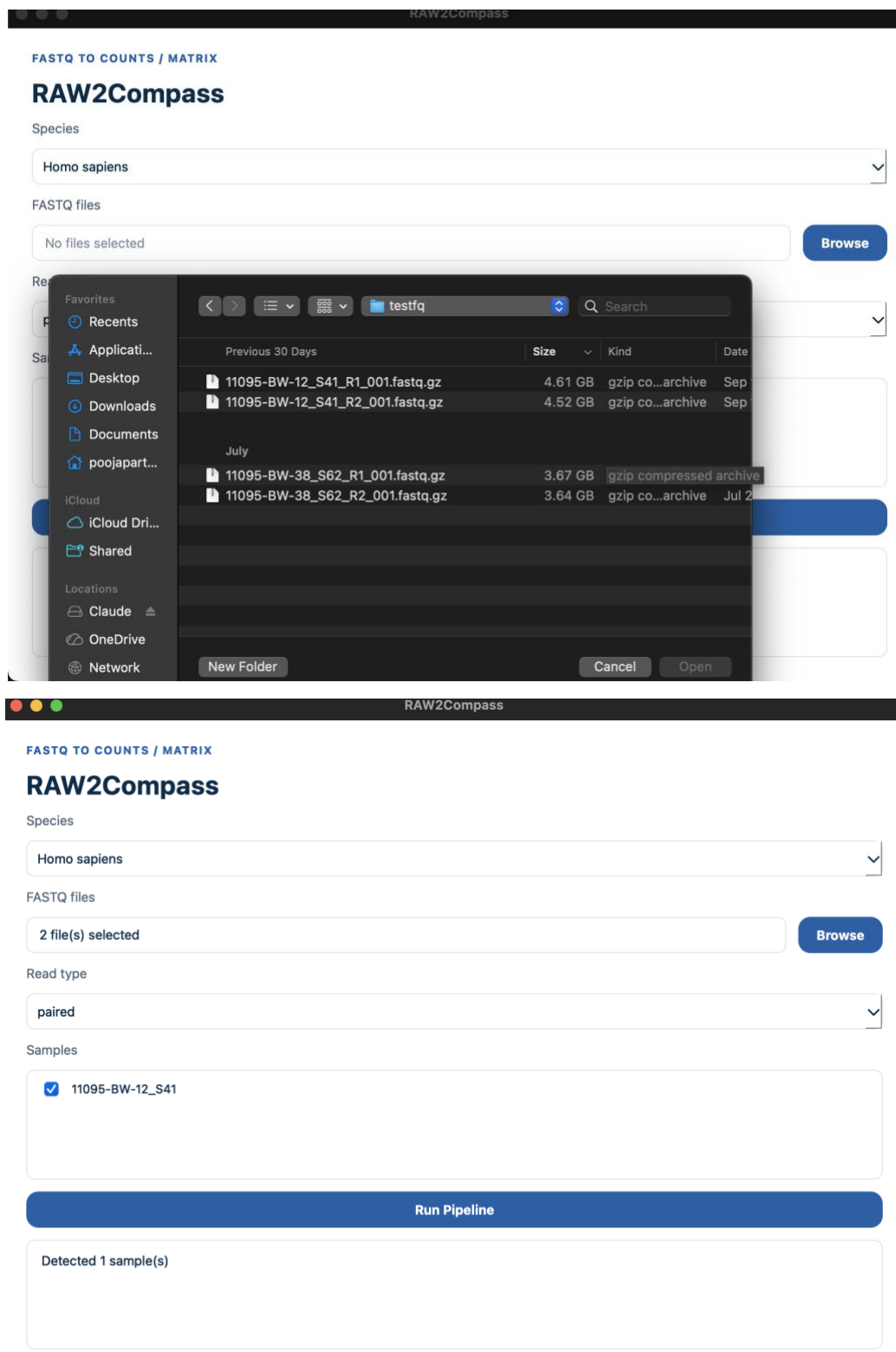
Run Pipeline

Screenshot 5: Species, read type, Browse button, and Run Pipeline button.

4. Prepare and select your FASTQ files

Before starting:

- Keep the FASTQ files together in a folder you can locate easily.
- The file picker accepts .fastq, .fastq.gz, .fq, and .fq.gz files.
- Make sure the files belong to the same species you will choose in the app.
- For paired-end reads, each sample must have matching forward and reverse files. The app recognizes these filename pairs:
 - _R1 and _R2
 - read1 and read2
 - forward and reverse
 - _1 and _2
- For single-end reads, each file is treated as its own sample.



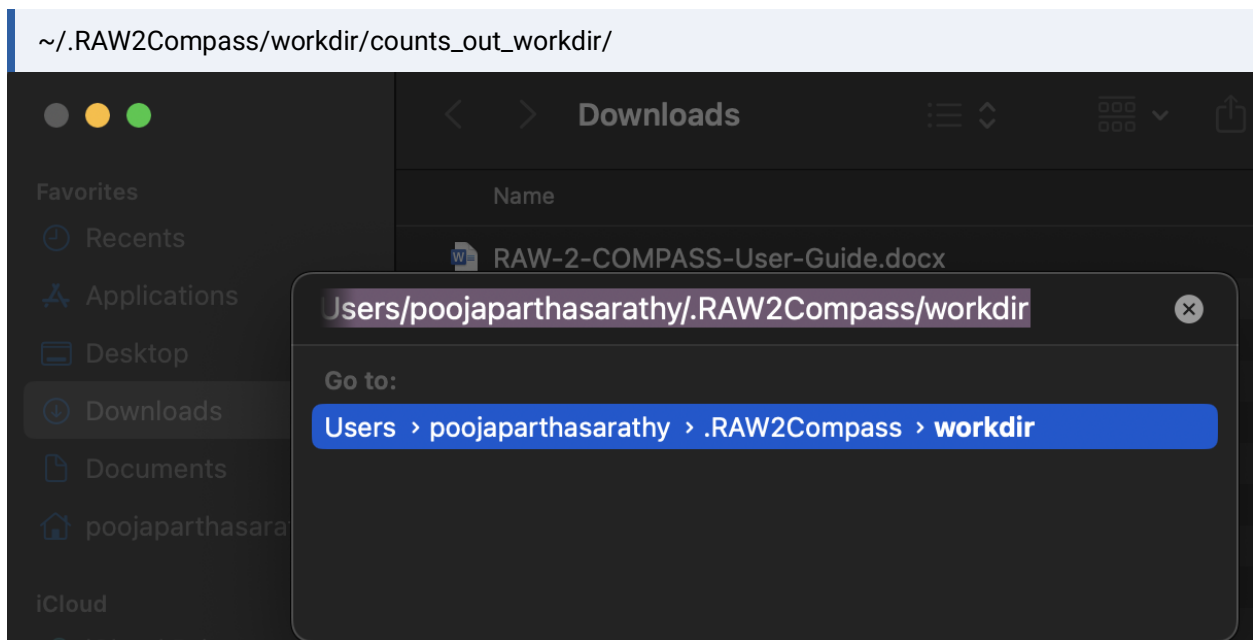
Screenshot 3a & 3b: FASTQ File selection.

5. Run the analysis

1. Confirm the correct species, read type, and checked samples.
2. Click Run Pipeline once.
3. Keep the app open while it runs. The log displays reference-building, download, indexing, and kallisto quantification messages.
4. The first run for a species may take longer because the reference must be downloaded and indexed.
5. When processing finishes, look for "Pipeline complete" and the output location in the log.
6. If the log shows "Error", note or copy the full message before contacting the app maintainer. Common causes include insufficient disk space, unavailable internet, unreadable FASTQ files, or mismatched paired-end filenames.

6. Find your results

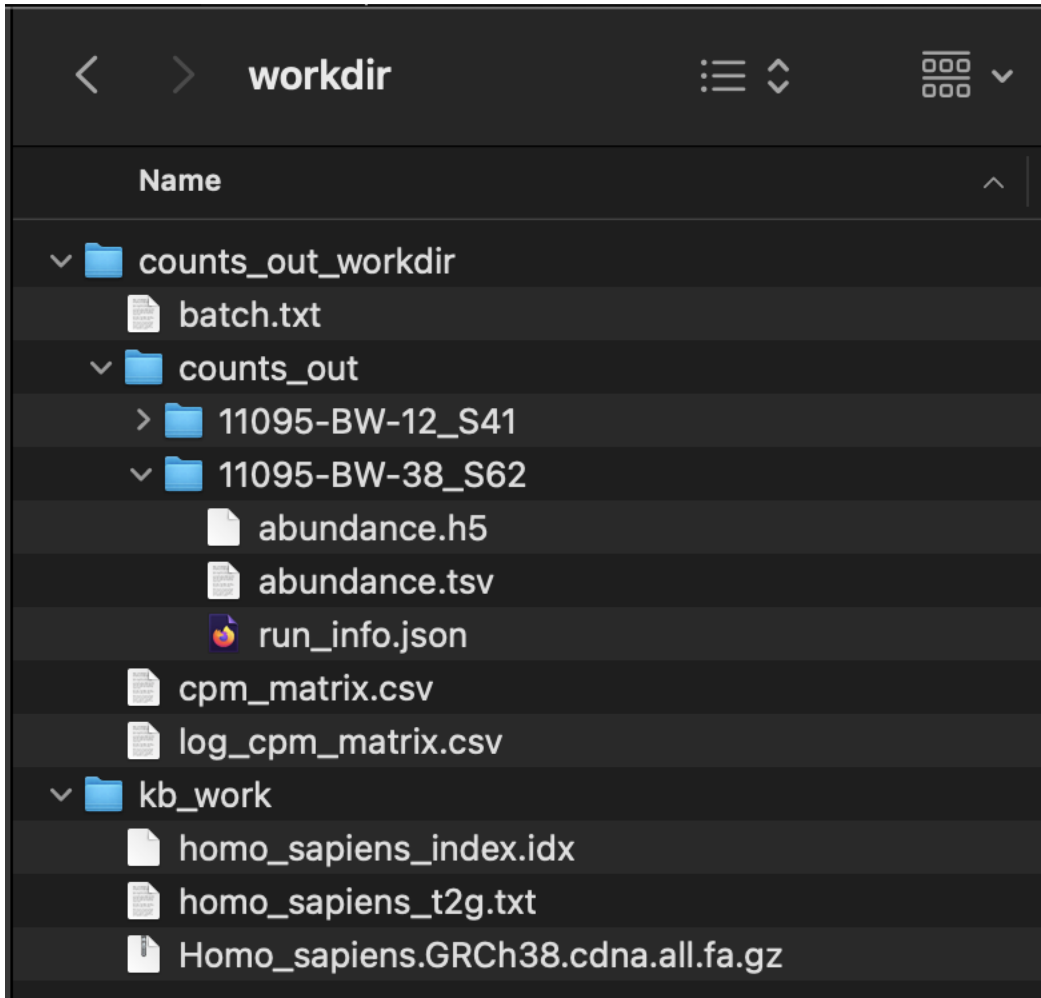
For the prebuilt macOS app, results are saved in your home folder at: (press Cmd + Shift + G)



The folder is hidden because its name starts with a period. To open it:

1. In Finder, choose Go > Go to Folder... from the menu bar.
2. Enter this path: `~/RAW2Compass/workdir/counts_out_workdir/`

3. Press Return.



The output folder contains:

- cpm_matrix.csv - gene-level counts normalized to counts per million (CPM).
- log_cpm_matrix.csv - $\log_2(\text{CPM} + 1)$ matrix.
- counts_out/ - one subfolder per sample, including kallisto's abundance.tsv quantification output.
- batch.txt - the sample names and FASTQ paths used for the run.

The matrix CSV files have gene symbols as rows and samples as columns. You can open them in spreadsheet software that supports CSV files. Keep the counts_out directory if you need the per-sample quantification files.

Reference FASTA files, transcript-to-gene mapping files, and kallisto indexes are stored separately in:

```
~/RAW2Compass/workdir/kb_work/
```

Do not delete these reference files if you want the app to reuse the species reference on a later run. (If you'd like to have a more recent edition of the transcript for next run, you may delete it for a latest reference to be pulled)

7. If you downloaded source code instead

The GitHub "Code > Download ZIP" option downloads source code, not the ready-to-open macOS app. To run source code, install Python 3.10 or newer and the project's dependencies in a local environment. The setup steps are in the repository README:

<https://github.com/poojabio/RAW-2-COMPASS>

The source-code workflow launches gui.py from a terminal and writes working files under the source project folder's workdir directory. It is different from the prebuilt app workflow described above.

Troubleshooting checklist

Problem	What to check
macOS says "Not Opened" or "Apple could not verify"	The app is unlicensed for now, so macOS quarantines it. Follow Step 2 to remove the quarantine flag with the xattr command, then open the app again.
App will not open	Confirm that the ZIP finished downloading and that you extracted the .app before opening it. If macOS says "Not Opened" or "Apple could not verify", follow Step 2. For any other security warning, see Step 3 and only open an app from a source you trust.
No samples detected	Confirm the read type was selected before Browse. For paired reads, select both files in each pair and check that their names match one of the supported patterns.
A paired file was skipped	Check that its matching mate is selected and has the expected R1/R2, read1/read2, forward/reverse, or _1/_2 naming.
The run fails during reference download	Check the internet connection and try again.
The run fails with a disk-space message	Free disk space and retry. FASTQ files, downloaded references, indexes, and per-sample results can require substantial storage.
Matrix files are missing	Check the final log lines for "Matrix export error". The per-sample counts may still exist in the counts_out folder even if matrix export failed.

When requesting help, include the macOS version, the app version if known, the selected species/read type, and the exact error text. Do not send human-subject FASTQ data unless authorized to share it.